Viral Hepatitis (A, B, C, D, E)

All key points for the exam + drug summary table

1. Overview of Hepatitis Viruses

Feature	HAV	HBV	HCV	HDV	HEV
Virus	RNA (picornavirus)	**DNA** (hepadnavirus)	RNA (flavivirus)	RNA (**defective** — needs HBsAg)	RNA
Transmission	**Fecal-oral** (food, water)	**Blood, sexual, perinatal**	**Blood** (**injection drug use** #1), perinatal, sexual (less)	Blood, sexual (only with HBV)	**Fecal-oral** (water), undercooked pork/game
Incubation	~28 days (15–50)	~90 days (60–150)	~45 days (14–180)	Similar to HBV	~40 days (15–60)
Chronic infection	**Never**	**Yes**: **90%** infants, 25–50% young children, **< 5% adults**	**Yes: 75–85%**	Yes (especially superinfection)	No (except in transplant or HIV patients)
Vaccine	**Yes**	**Yes**	**No**	**HBV vaccine** prevents it	Not available in US (China only)
Treatment	Supportive	Antivirals (suppression)	**DAAs — cure > 95%**	Peginterferon	Supportive (ribavirin in chronic)

Memory aid: "**Vowels hit your Bowels**" → **A and E** are **fecal-oral**, acute only (no chronic). **B, C, D** are **blood-borne** and can become **chronic** → **cirrhosis** and **hepatocellular carcinoma (HCC)**.

2. Clinical Presentation and Labs

Phase	Features
Prodrome	**Fatigue**, malaise, **anorexia**, **nausea**, vomiting, low-grade fever, **RUQ pain**, arthralgia
Icteric phase	**Jaundice**, **dark urine**, **pale (clay-colored) stools**, pruritus, hepatomegaly
Many patients	Asymptomatic (especially children with HAV, and most HCV)
Acute liver failure (fulminant)	**Hepatic encephalopathy** + **INR ≥ 1.5** in a patient without prior liver disease → **liver transplant** evaluation
Chronic hepatitis	Often silent for years → **cirrhosis** (ascites, varices, encephalopathy) and **HCC**

- **Labs: ALT > AST** (often **> 1,000 U/L** in acute viral hepatitis) • ↑ bilirubin • **INR** and **albumin** reflect liver **synthetic function**.
- **Alcoholic hepatitis** → **AST:ALT > 2:1** (contrast with viral).
- **Fibrosis assessment: FIB-4**, **APRI**, **elastography** (FibroScan), biopsy (rarely).

3. Hepatitis A (HAV)

- **Fecal-oral** spread: contaminated food/water, travel, close contacts, **MSM**, **people who use drugs**, **homeless**, daycare.
- **Self-limited**, **no chronic** infection; **lifelong immunity** after infection.
- **Diagnosis: IgM anti-HAV** = acute infection • IgG anti-HAV = past infection or vaccination (immune).
- **Treatment: supportive** (rest, hydration) • **avoid alcohol** and hepatotoxic drugs • acetaminophen at reduced doses.
- More severe in **older adults** and patients with **chronic liver disease**.

Hepatitis A vaccine (inactivated)

Product	Dose	Schedule
Havrix	Children 720 ELU • **Adults (≥ 19) 1,440 ELU** IM	**2 doses: 0 and 6–12 months**
Vaqta	Children 25 units • **Adults 50 units** IM	**2 doses: 0 and 6–18 months**
Twinrix (HAV + HBV)	Adults ≥ 18 (720 ELU HAV + 20 mcg HBsAg)	**3 doses: 0, 1, 6 months** • Accelerated: 0, 7, 21–30 days + booster at 12 months

Who should get the HAV vaccine

- **All children at 12–23 months** (routine) and catch-up for all children/adolescents 2–18 years.
- Adults at risk: **international travelers** (to endemic areas), **MSM**, **people who use drugs**, **homeless**, **chronic liver disease** (including **HBV and HCV**), **HIV**, occupational risk, close contact with international adoptees, pregnancy with risk.
- **Travelers:** give the first dose **as soon as travel is planned** (one dose gives protection) • Infants **6–11 months** traveling: one dose (does not count toward the routine series).

Post-exposure prophylaxis (PEP) — within 2 weeks of exposure

Person	PEP
Healthy, ≥ 12 months	**HAV vaccine** (single dose, then complete the series)
Adults > 40 years	**Vaccine** (IG may be added based on risk)
Immunocompromised or **chronic liver disease**	**Vaccine + immune globulin (IG)**
Infants < 12 months or vaccine contraindication	**Immune globulin (GamaSTAN) 0.1 mL/kg IM**

IG for travel (if vaccine not possible): **0.1 mL/kg** (up to 1 month of travel), **0.2 mL/kg** (up to 2 months; repeat every 2 months).

4. Hepatitis B (HBV)

Transmission and risk groups

- **Blood**, **sexual contact**, **perinatal** (mother to infant — most common worldwide), shared needles, razors, toothbrushes.
- **Not** spread by hugging, kissing, sharing food/utensils, coughing, or **breastfeeding**.
- **Risk of chronic infection** depends on age: **~90% of infected newborns** vs **< 5% of adults** → this is why **infant vaccination** and **perinatal prevention** are crucial.
- **Screen** all adults **at least once** (HBsAg, anti-HBc, anti-HBs) and **every pregnancy** (HBsAg).

HBV serologic markers

Marker	Meaning
HBsAg (surface antigen)	**Current infection** (acute or chronic) • **Positive > 6 months = chronic HBV**
Anti-HBs (surface antibody)	**Immunity** — from **vaccination** or **recovery** • **≥ 10 mIU/mL** = protective
Anti-HBc total (core antibody)	**Exposure** to the virus (past or current) • **NOT produced by vaccination**
IgM anti-HBc	**Acute (recent) infection** (< 6 months)
HBeAg (e antigen)	**Active viral replication**, **high infectivity**
Anti-HBe	Lower replication (HBeAg seroconversion)
HBV DNA	**Viral load** (replication) • Used to decide treatment and monitor response

Interpreting HBV serology (high-yield)

HBsAg	Anti-HBc	IgM anti-HBc	Anti-HBs	Interpretation
–	–	–	–	**Susceptible** (never infected, not immune) → **vaccinate**
–	–	–	+	**Immune from vaccination**
–	+	–	+	**Immune from natural infection** (resolved/recovered)
+	+	+	–	**Acute infection**
+	+	–	–	**Chronic infection**
–	+	–	–	**Isolated anti-HBc**: resolved infection (most common), false positive, or occult HBV → check HBV DNA

Phases of chronic HBV (AASLD)

Phase	HBeAg	HBV DNA	ALT	Treat?
Immune-tolerant	+	Very high (> 1 million)	**Normal**	Usually **no** (monitor)
HBeAg-positive immune-active	+	High (> 20,000)	**Elevated**	**Yes**
Inactive carrier	–	Low (< 2,000)	**Normal**	**No** (monitor)
HBeAg-negative immune-active (reactivation)	–	≥ 2,000	**Elevated**	**Yes**

Who to treat (AASLD)

- **Immune-active chronic HBV: ALT > 2 × ULN** (ULN: **35 U/L men**, **25 U/L women**) or significant liver inflammation/fibrosis, **plus** HBV DNA **> 20,000 IU/mL (HBeAg+)** or **> 2,000 IU/mL (HBeAg−)**.
- **Compensated cirrhosis** with HBV DNA > 2,000 IU/mL (many experts: any detectable DNA).
- **Decompensated cirrhosis** with **any detectable HBV DNA** (refer for transplant).
- **Acute liver failure**, severe acute HBV, **HCC**, extrahepatic manifestations.
- **Pregnancy** with **HBV DNA > 200,000 IU/mL** → **tenofovir (TDF)** starting at **28–32 weeks** to prevent perinatal transmission.
- **Prophylaxis** before **immunosuppression or chemotherapy** (see reactivation).
- Note: newer guidelines (e.g., WHO 2024) recommend **broader** treatment criteria.

Antiviral therapy for HBV

Drug	Dose	Key points
Entecavir (Baraclude) — **preferred**	**0.5 mg daily** (treatment-naïve) • **1 mg daily** (lamivudine-resistant or **decompensated cirrhosis**)	**Take on an empty stomach** (2 h after and 2 h before a meal) • **High barrier to resistance** • **Avoid if lamivudine resistance** (cross-resistance) • Renal dose adjustment (CrCl < 50) • Limited pregnancy data
Tenofovir disoproxil fumarate (TDF) (Viread) — **preferred**	**300 mg daily**	**Nephrotoxicity** (↑ SCr, **Fanconi syndrome**, hypophosphatemia) • **↓ bone density** • **Preferred in pregnancy** • Renal adjustment (CrCl < 50) • Works against lamivudine-resistant HBV
Tenofovir alafenamide (TAF) (Vemlidy) — **preferred**	**25 mg daily with food**	**Less kidney and bone toxicity** than TDF (preferred if CKD, osteoporosis, elderly) • May ↑ lipids • Not recommended if CrCl < 15 and not on dialysis • **P-gp inducers** (rifampin, carbamazepine, phenytoin, St. John's wort) ↓ levels → avoid
Peginterferon alfa-2a (Pegasys)	**180 mcg SC once weekly × 48 weeks**	**Finite duration**, no resistance • Best in young, HBeAg+, high ALT, low DNA • **Contraindicated: decompensated cirrhosis**, **pregnancy**, autoimmune disease, **severe depression/psychiatric illness**, uncontrolled seizures • ADRs: **flu-like symptoms**, **depression**, cytopenias, **thyroid dysfunction**, fatigue, hair loss
Lamivudine (Epivir-HBV), **adefovir** (Hepsera)	Lamivudine 100 mg daily • Adefovir 10 mg daily	**Not preferred**: **lamivudine → high resistance (YMDD)** • **adefovir → nephrotoxic**, weak • (Telbivudine discontinued)

Boxed warnings (nucleos(t)ide analogs)

- **Severe acute exacerbation of hepatitis B** after **stopping** therapy → **do not stop** without medical advice; monitor LFTs for several months after stopping.
- **HIV resistance** if used in patients with **untreated HIV** → **test for HIV before starting** (HIV/HBV coinfection: use a full HIV regimen containing **tenofovir + emtricitabine/lamivudine**).
- **Lactic acidosis** and severe hepatomegaly with steatosis (entecavir, older agents).

Duration and endpoints

- **HBeAg-positive, no cirrhosis:** until **HBeAg seroconversion** (anti-HBe), then **≥ 12 months consolidation** (or until HBsAg loss).
- **HBeAg-negative:** usually **indefinite** (until **HBsAg loss**).
- **Cirrhosis: indefinite** (lifelong).
- **Ideal endpoint** = **HBsAg loss** (functional cure; rare). Antivirals **suppress** HBV but rarely **cure** it (cccDNA persists).

Monitoring

- **ALT and HBV DNA every 3–6 months** • HBeAg/anti-HBe • HBsAg yearly.
- **SCr, phosphate** (TDF, adefovir) • bone density (TDF) • lipids (TAF).
- **Adherence** (resistance, flares).
- **HCC surveillance: ultrasound ± AFP every 6 months** in **cirrhosis** and high-risk groups (Asian men > 40, Asian women > 50, African-born, family history of HCC) — even if HBV is suppressed.

HBV vaccines (recombinant, inactivated)

Vaccine	Adult dose	Schedule
Engerix-B	**20 mcg (1 mL)** IM • **Dialysis: 40 mcg**	**0, 1, 6 months** (dialysis: 0, 1, 2, 6)
Recombivax HB	10 mcg IM • **Dialysis formulation 40 mcg**	**0, 1, 6 months** • Adolescents 11–15: 2-dose option (0, 4–6 months)
Heplisav-B (CpG adjuvant)	20 mcg IM (adults ≥ 18)	**2 doses: 0 and 1 month** (faster, higher response)
Twinrix (HAV + HBV)	Adults ≥ 18	0, 1, 6 months

- **Infants: birth dose within 24 hours** (medically stable, ≥ 2 kg), complete series by 6–18 months.
- **Adults: universal vaccination for ages 19–59**; **≥ 60** with risk factors (others may receive it).
- **Post-vaccination testing** (anti-HBs 1–2 months after the series): **healthcare workers**, **dialysis**, **HIV**/immunocompromised, **sex partners of HBsAg+ people**, **infants of HBsAg+ mothers** (at **9–12 months**).
- **Non-responder** (anti-HBs < 10): **repeat the full series once**, then retest • still negative → check HBsAg (may be infected).
- **Contraindication:** severe allergy to **yeast** (Engerix-B, Recombivax HB) or prior dose.

Preventing perinatal transmission

Situation	Action
Mother HBsAg-positive	Infant: **HBIG 0.5 mL IM + HBV vaccine within 12 hours of birth** (different sites) • Complete the series • **Test infant** (HBsAg + anti-HBs) at **9–12 months**
Mother HBV DNA > 200,000 IU/mL	**Tenofovir (TDF)** from **28–32 weeks** until delivery (to ~3 months postpartum)
Mother HBsAg status unknown	Vaccine within 12 hours • test mother • HBIG within 7 days if mother is positive (within 12 h if infant < 2 kg)
Breastfeeding	**Allowed** (after HBIG + vaccine), including on tenofovir

Post-exposure prophylaxis (needlestick, sexual exposure)

Exposed person	Source HBsAg-positive (or unknown, high risk)
Unvaccinated or incomplete series	**HBIG 0.06 mL/kg IM + start HBV vaccine** (ideally within 24 h; ≤ 7 days needlestick, ≤ 14 days sexual)
Vaccinated, known responder (anti-HBs ≥ 10)	**No treatment**
Vaccinated, known non-responder	**HBIG × 2 doses** (1 month apart), or HBIG × 1 + revaccinate
Vaccinated, response unknown	Test anti-HBs now → ≥ 10: nothing • < 10: HBIG + vaccine booster

HBV reactivation with immunosuppression (high-yield)

- **Screen** (**HBsAg, anti-HBc**, ± anti-HBs) **before** chemotherapy, **rituximab** (anti-CD20), **anti-TNF**, high-dose steroids, transplant immunosuppression, JAK inhibitors, and **before HCV DAA therapy**.
- **HBsAg-positive** → **prophylaxis** with **entecavir, TDF, or TAF** (not lamivudine) — start before or with immunosuppression and continue **6–12 months after** (**12–18 months after rituximab**).
- **HBsAg-negative / anti-HBc-positive** → **prophylaxis** if **high risk** (e.g., **rituximab**, stem cell transplant); otherwise monitor HBsAg/HBV DNA.
- **Rituximab**, ofatumumab, obinutuzumab carry a **boxed warning** for HBV reactivation.

5. Hepatitis D (HDV, delta)

- **Defective virus** → needs **HBsAg** to replicate → occurs **only with HBV**.
- **Coinfection** (HBV + HDV at the same time) → usually resolves • **Superinfection** (HDV in a chronic HBV carrier) → **more severe**, **chronic**, rapid progression to **cirrhosis**.
- **Test for HDV (anti-HDV, then HDV RNA)** in HBsAg-positive patients (especially high-risk; many guidelines now advise testing all).
- **Treatment: peginterferon alfa × 48 weeks** (limited efficacy) • **Bulevirtide** (entry inhibitor) approved in Europe, not the US • Nucleos(t)ide analogs do **not** work against HDV (used only if HBV DNA is high).
- **Prevention: HBV vaccination**.

6. Hepatitis C (HCV)

Epidemiology and natural history

- **Transmission: injection drug use** (most common), blood transfusion/organ transplant **before 1992**, **hemodialysis**, needlesticks, **perinatal** (~5%), sexual (higher in **MSM** and **HIV**), unsterile tattoos.
- **Chronic infection in 75–85% → cirrhosis** in ~20% over 20–30 years → **HCC** (alcohol, HIV, HBV, obesity speed progression).
- **Genotypes 1–6**: genotype 1 most common in the US; **genotype 4** common in the **Middle East** and Egypt; genotype 3 progresses faster.
- **No vaccine** and **no post-exposure prophylaxis**.
- **Extrahepatic manifestations: mixed cryoglobulinemia**, **membranoproliferative glomerulonephritis**, **porphyria cutanea tarda**, lichen planus, **type 2 diabetes**, non-Hodgkin lymphoma.

Screening and diagnosis

- **Screen all adults ≥ 18 at least once** (CDC) and **every pregnancy** • **Periodic** testing with ongoing risk (e.g., injection drug use).
- **Step 1: HCV antibody** (anti-HCV) → positive = exposure (past or current).
- **Step 2: HCV RNA** → **positive = current infection** • negative = cleared infection (~25% clear spontaneously) or false-positive antibody.
- **Antibody stays positive after cure** → use **HCV RNA** to diagnose **reinfection**.
- After a **needlestick**: baseline testing of source and exposed person; HCV RNA at 3–6 weeks; antibody at ≥ 6 months.

Pretreatment evaluation

- **Everyone with chronic HCV should be treated** (except very short life expectancy).
- **HCV RNA** (viral load) • **Genotype** (needed for cirrhosis or prior treatment; not required for simplified pangenotypic treatment).
- **Fibrosis/cirrhosis assessment**: FIB-4, elastography • **Child-Pugh score** if cirrhosis (decompensated = Child-Pugh B or C).
- **HBV testing** (HBsAg, anti-HBc, anti-HBs) — **boxed warning: HBV reactivation** during DAA therapy • HIV test • pregnancy test.
- **Drug interaction review** (very important) • vaccinate against **HAV and HBV** if not immune.

Direct-acting antivirals (DAAs)

Class	Suffix	Drugs
NS3/4A protease inhibitors	**-previr**	**Glecaprevir**, **voxilaprevir**, **grazoprevir** • **Contraindicated in decompensated cirrhosis (Child-Pugh B/C)** → risk of liver failure
NS5A inhibitors	**-asvir**	**Velpatasvir**, **ledipasvir**, **pibrentasvir**, **elbasvir** • Absorption of velpatasvir and ledipasvir is **pH-dependent** (acid suppressant interactions)
NS5B polymerase inhibitor (nucleotide)	**-buvir**	**Sofosbuvir** • **High barrier to resistance** • **Amiodarone → severe bradycardia** (avoid)

Recommended regimens

Regimen	Dose	Duration / use
Glecaprevir/pibrentasvir (Mavyret)	**3 tablets (100/40 mg each) once daily with food**	**8 weeks** — treatment-naïve, **with or without compensated cirrhosis**, all genotypes • **No renal adjustment** • **Not for decompensated cirrhosis**
Sofosbuvir/velpatasvir (Epclusa)	**400/100 mg once daily**	**12 weeks** — treatment-naïve, with or without compensated cirrhosis, all genotypes (GT3 with cirrhosis: check **NS5A RAS Y93H**) • **Decompensated cirrhosis:** + **weight-based ribavirin × 12 weeks** (or 24 weeks without ribavirin)
Ledipasvir/sofosbuvir (Harvoni)	**90/400 mg once daily**	Genotypes 1, 4, 5, 6 • **8 weeks** if treatment-naïve, no cirrhosis, HCV RNA < 6 million • otherwise 12 weeks
Elbasvir/grazoprevir (Zepatier)	**50/100 mg once daily**	Genotypes 1, 4 • 12 weeks • GT1a: test **NS5A RAS** first • Not for decompensated cirrhosis
Sofosbuvir/velpatasvir/voxilaprevir (Vosevi)	**400/100/100 mg once daily with food**	**12 weeks** — **retreatment** after DAA failure (including NS5A inhibitor) • Not for decompensated cirrhosis
Ribavirin (add-on only)	**Weight-based: 1,000 mg/day (< 75 kg) or 1,200 mg/day (≥ 75 kg)**, divided BID with food	Decompensated cirrhosis, some retreatment • **Never as monotherapy**

Simplified treatment (treatment-naïve, no decompensated cirrhosis, no HBV/HIV coinfection, not pregnant, no prior treatment): **Mavyret × 8 weeks** or **Epclusa × 12 weeks → no genotype needed**, minimal monitoring.

Monitoring and cure

- **HCV RNA at baseline** and **12 weeks after the end of treatment → undetectable = sustained virologic response (SVR12) = CURE** (> 95%).
- During treatment: adherence, drug interactions, LFTs (and glucose in diabetics — **hypoglycemia** as HCV clears; **INR** changes on warfarin).
- **After cure: no immunity → reinfection is possible** (counsel, harm reduction) • patients with **cirrhosis** still need **HCC surveillance every 6 months** and variceal screening.

DAA drug interactions (high-yield)

Interacting drug	Interaction / action
Amiodarone	**Sofosbuvir-containing** regimens → **severe symptomatic bradycardia**, deaths → **avoid** (if unavoidable, cardiac monitoring)
Acid suppressants	↓ **velpatasvir/ledipasvir** absorption • **PPIs:** avoid; if needed with Epclusa → take Epclusa **with food 4 h before omeprazole 20 mg** • **H2RAs:** at the same time or 12 h apart (max famotidine 40 mg BID) • **Antacids:** separate by **4 hours**
Strong P-gp / CYP inducers	**Rifampin**, **carbamazepine**, **phenytoin**, phenobarbital, **St. John's wort** → ↓ DAA levels → **contraindicated**
Statins	↑ statin levels (myopathy) • **Mavyret:** avoid **atorvastatin, simvastatin, lovastatin** • **Harvoni:** avoid **rosuvastatin** • **Epclusa:** rosuvastatin ≤ 10 mg
Ethinyl estradiol	**Mavyret** (and Zepatier) → **ALT elevations** → avoid estrogen-containing contraceptives (use progestin-only or IUD)
HIV drugs	**Mavyret:** contraindicated with **atazanavir**; avoid efavirenz, ritonavir-boosted PIs • **Tenofovir (TDF)** levels ↑ with ledipasvir/velpatasvir → monitor renal function • **Zepatier:** avoid efavirenz, boosted PIs
Others	**Dabigatran**, **digoxin**, **cyclosporine** (↑ levels) • **Warfarin** (INR changes as liver function improves) • **Diabetes drugs** (hypoglycemia risk)

Ribavirin

- **Boxed warnings: hemolytic anemia** (can worsen cardiac disease) • **teratogenic** (also via male partner) → **2 reliable forms of contraception** during therapy and for **6 months after** (both partners) • **not effective as monotherapy**.
- Other ADRs: fatigue, cough, dyspnea, rash, pruritus, insomnia, hyperuricemia.
- **Renal dose adjustment** • DDIs: **didanosine** (contraindicated: lactic acidosis, pancreatitis), **azathioprine** (myelotoxicity), **zidovudine** (anemia).

Special populations

Population	Key points
Decompensated cirrhosis	**No protease inhibitors** (-previr) • **Sofosbuvir/velpatasvir + ribavirin** (or ledipasvir/sofosbuvir + RBV) • Refer to transplant center
CKD / dialysis	**Mavyret** or **Epclusa** — **no dose adjustment** (sofosbuvir is now acceptable at any eGFR)
HBV coinfection	Treat/monitor HBV — **HBV reactivation** risk during DAAs (prophylaxis if HBsAg+)
HIV coinfection	Same regimens and durations • check antiretroviral DDIs
Pregnancy	Test all pregnant women • DAAs **not approved** but may be considered case by case • **Ribavirin contraindicated** • Breastfeeding allowed (unless nipples cracked/bleeding) • Test infant: HCV RNA at 2–6 months or antibody at ≥ 18 months
Children	DAAs approved from **age 3** (weight-based)
Acute HCV	**Treat promptly** with the same regimens (no need to wait for spontaneous clearance)
People who inject drugs	**Treat** (active use is not a contraindication) + harm reduction (needle exchange, opioid agonist therapy)

7. Hepatitis E (HEV)

- **Fecal-oral** (contaminated **water**; outbreaks in developing countries) • zoonotic: **undercooked pork**, wild boar, deer.
- Usually **self-limited** (like HAV) → **supportive care**.
- **Pregnancy (3rd trimester):** risk of **fulminant hepatitis** with **high mortality (~20–25%)** — high-yield.
- **Chronic HEV** in **immunocompromised** (solid organ transplant, HIV) → **reduce immunosuppression** ± **ribavirin for ~3 months**.
- **Diagnosis:** IgM anti-HEV, HEV RNA • **Vaccine** (Hecolin) available only in China.

8. General Management and Counseling

Topic	Key points
Acute hepatitis	**Supportive** • Rest, hydration, nutrition • Hospitalize if encephalopathy, ↑ INR, inability to eat/drink
Alcohol	**Avoid** (accelerates liver damage)
Acetaminophen	Can be used at **reduced doses** (generally **≤ 2 g/day** in chronic liver disease/cirrhosis) — preferred over NSAIDs
NSAIDs	**Avoid in cirrhosis** (renal failure, GI bleeding, ascites)
Hepatotoxic drugs / herbals	Avoid unnecessary hepatotoxins and herbal supplements (e.g., kava)
Vaccination	Patients with chronic HBV or HCV → **HAV vaccine** (and HBV vaccine in HCV patients) if not immune • pneumococcal, influenza
Prevent spread (HBV/HCV)	**Do not share** razors, toothbrushes, nail clippers, needles • cover cuts • safe sex (HBV; HCV in high-risk) • do not donate blood/organs/semen • **casual contact is safe**

Topic	Key points
Prevent spread (HAV/HEV)	**Hand washing**, safe food and water, vaccinate contacts (HAV)
Liver cancer surveillance	**Ultrasound ± AFP every 6 months** in cirrhosis (any cause) and high-risk HBV

Exam Traps and Frequently Tested Points

1. **A and E** = **fecal-oral**, acute only • **B, C, D** = blood-borne, chronic.
2. **HBV chronicity**: **90% newborns** vs **< 5% adults** • **HCV chronicity**: **75–85%**.
3. **HEV in pregnancy** → fulminant hepatitis, high mortality.
4. **HDV** needs **HBsAg**; superinfection is worse; **HBV vaccine prevents HDV**.
5. **HCV has no vaccine** and no PEP • **HAV and HBV have vaccines**.
6. **IgM anti-HAV** = acute HAV • **IgM anti-HBc** = acute HBV.
7. **HBsAg > 6 months** = chronic HBV • **anti-HBs alone** = vaccinated • **anti-HBc + anti-HBs** = recovered.
8. **Anti-HBc is NOT produced by vaccination** (only infection).
9. **HBeAg** = high replication and infectivity.
10. **HAV vaccine:** 2 doses (0, 6–12/18 months) • PEP within **2 weeks**: vaccine (healthy ≥ 1 year), **IG 0.1 mL/kg** (< 12 months), both if immunocompromised/chronic liver disease.
11. **HBV vaccine:** Engerix-B/Recombivax 0, 1, 6 months • **Heplisav-B 2 doses (0, 1 month)** • **dialysis 40 mcg** • **birth dose within 24 h** • protective anti-HBs **≥ 10 mIU/mL**.
12. **Infant of HBsAg+ mother: HBIG + vaccine within 12 hours** • test at 9–12 months • **breastfeeding OK**.
13. **Mother HBV DNA > 200,000** → **TDF from 28–32 weeks**.
14. **HBV first-line drugs: entecavir**, **TDF**, **TAF** (and peginterferon) • **avoid lamivudine** (resistance) and adefovir.
15. **Entecavir: empty stomach** • 0.5 mg (naïve), **1 mg** (lamivudine-resistant, decompensated) • avoid if lamivudine resistance.
16. **TDF: kidney (Fanconi) and bone** toxicity • **preferred in pregnancy** • **TAF**: less kidney/bone toxicity, take with food.
17. **Stopping HBV antivirals** → **severe hepatitis flare** (boxed) • **test HIV** before starting (resistance).
18. **Peginterferon:** finite, no resistance • **contraindicated in decompensated cirrhosis, pregnancy, severe depression** • flu-like symptoms, depression, thyroid problems.
19. **Screen HBsAg + anti-HBc before** chemotherapy, **rituximab**, anti-TNF, and **HCV DAAs** → prophylaxis with entecavir/tenofovir.
20. **HCV screening:** all adults once + every pregnancy • **antibody → confirm with HCV RNA**.
21. **HCV antibody stays positive after cure** → use RNA for reinfection.
22. **Cure = SVR12** (undetectable HCV RNA 12 weeks after treatment).
23. **Mavyret: 3 tablets daily with food × 8 weeks** • **Epclusa: 1 tablet daily × 12 weeks** — both pangenotypic.
24. **Protease inhibitors (-previr) contraindicated in decompensated cirrhosis** → use **sofosbuvir/velpatasvir + ribavirin**.
25. **Sofosbuvir + amiodarone** → severe bradycardia.
26. **Velpatasvir/ledipasvir + PPIs/H2RAs/antacids** → ↓ absorption (special timing rules).
27. **Rifampin, carbamazepine, phenytoin, St. John's wort + DAAs** → contraindicated.
28. **Mavyret + ethinyl estradiol** → ALT elevation; **Mavyret + atorvastatin/simvastatin** → avoid.
29. **Ribavirin:** hemolytic anemia + **teratogenic** (2 contraceptives, 6 months after, both partners) • never monotherapy.
30. **After HCV cure:** reinfection possible; cirrhosis still needs **HCC surveillance every 6 months**.
31. **Chronic liver disease:** acetaminophen ≤ 2 g/day, avoid NSAIDs and alcohol, vaccinate HAV/HBV.
32. **ALT > AST** in viral hepatitis; **AST:ALT > 2** in alcoholic hepatitis.

Note: Treatment thresholds and regimens are updated often (AASLD/IDSA, WHO, CDC). **Use the numbers in the reference you are studying for the exam.**

Drug Summary Table: Doses, Key ADRs and DDIs

Drug	Dose	Key ADRs	Key DDIs / special notes
Hepatitis A vaccine (Havrix, Vaqta)	Adults: Havrix 1,440 ELU / Vaqta 50 U IM • 0 and 6–12 (Havrix) or 6–18 (Vaqta) months	Injection site pain, headache	Inactivated (safe in immunocompromised and pregnancy) • PEP within 2 weeks
Immune globulin (GamaSTAN)	HAV PEP 0.1 mL/kg IM • travel 0.1–0.2 mL/kg	Injection pain, rare anaphylaxis (IgA deficiency)	Delays **MMR/varicella** live vaccines (wait ≥ 3–6 months)
Twinrix (HAV + HBV)	0, 1, 6 months (or 0, 7, 21–30 days + 12 months)	Injection site reactions	Adults ≥ 18
Engerix-B	20 mcg IM at 0, 1, 6 months • dialysis 40 mcg at 0, 1, 2, 6	Injection site pain, fever	Contraindicated in **yeast** allergy
Recombivax HB	10 mcg IM at 0, 1, 6 months • dialysis 40 mcg	Same	Yeast allergy
Heplisav-B	20 mcg IM at 0 and 1 month	Injection site reactions, headache	Adults ≥ 18 • faster protection
HBIG	Infant 0.5 mL IM within 12 h • PEP 0.06 mL/kg IM	Injection site pain	Give with vaccine at a different site
Entecavir (Baraclude)	0.5 mg daily (naïve) • 1 mg daily (lamivudine-resistant/decompensated)	Headache, fatigue • **boxed: hepatitis flare on stopping, lactic acidosis, HIV resistance**	**Empty stomach** • renal adjustment (CrCl < 50) • avoid if lamivudine resistance • test HIV first
Tenofovir DF (Viread)	300 mg daily	**Nephrotoxicity (Fanconi)**, **↓ bone density**, GI • boxed: flare on stopping	Renal adjustment (CrCl < 50) • avoid other nephrotoxins • ↑ levels with ledipasvir/velpatasvir • **preferred in pregnancy**
Tenofovir AF (Vemlidy)	25 mg daily with food	Headache, nausea, ↑ lipids (less kidney/bone toxicity) • boxed: flare on stopping	**P-gp inducers** (rifampin, carbamazepine, phenytoin, St. John's wort) → avoid • not if CrCl < 15 without dialysis
Peginterferon alfa-2a (Pegasys)	180 mcg SC weekly × 48 weeks	**Flu-like symptoms**, **depression/suicidality**, cytopenias, **thyroid dysfunction**, alopecia • boxed: neuropsychiatric, autoimmune, ischemic, infectious disorders	**CI: decompensated cirrhosis, pregnancy**, autoimmune disease • ↑ theophylline • myelosuppression with zidovudine
Lamivudine (Epivir-HBV)	100 mg daily (HBV)	Well tolerated; **high resistance**	Not preferred • HIV dose differs (300 mg)
Adefovir (Hepsera)	10 mg daily	**Nephrotoxicity**	Not preferred
Glecaprevir/pibrentasvir (Mavyret)	3 tablets once daily with food × 8 weeks	Headache, fatigue, nausea • boxed: HBV reactivation	**CI: decompensated cirrhosis**, **atazanavir**, **rifampin** • avoid **ethinyl estradiol**, atorvastatin, simvastatin, carbamazepine, efavirenz • ↑ dabigatran, digoxin
Sofosbuvir/velpatasvir (Epclusa)	1 tablet (400/100 mg) daily × 12 weeks	Headache, fatigue, nausea • boxed: HBV reactivation	**Amiodarone** (bradycardia) • **acid suppressants** (timing rules) • **inducers CI** • rosuvastatin ≤ 10 mg • ↑ TDF, digoxin
Ledipasvir/sofosbuvir (Harvoni)	1 tablet (90/400 mg) daily × 8–12 weeks	Headache, fatigue • boxed: HBV reactivation	**Amiodarone** • acid suppressants • **inducers CI** • **rosuvastatin** avoid • ↑ TDF

Drug	Dose	Key ADRs	Key DDIs / special notes
Elbasvir/grazoprevir (Zepatier)	1 tablet (50/100 mg) daily × 12 weeks	Fatigue, headache, ↑ **ALT** • boxed: HBV reactivation	**CI: decompensated cirrhosis**, **OATP1B inhibitors** (cyclosporine, atazanavir, darunavir), strong CYP3A inducers, efavirenz
Sofosbuvir/velpatasvir/voxilaprevir (Vosevi)	1 tablet daily with food × 12 weeks	Headache, diarrhea, nausea • boxed: HBV reactivation	**Retreatment** • CI with rifampin, decompensated cirrhosis • amiodarone • rosuvastatin CI
Ribavirin	1,000 mg/day (< 75 kg) or 1,200 mg/day (≥ 75 kg), divided BID with food	**Boxed: hemolytic anemia, teratogenicity** • cough, dyspnea, rash, fatigue	2 contraceptives (both partners) during + 6 months after • CI with **didanosine** • ↑ myelotoxicity with azathioprine • anemia with zidovudine • renal adjustment

Good luck on your exam!